

SYNOPSIS

The 6454gt UK registered container ship *Cepheus J*, and the 4955gt Maltese registered general cargo ship *Ileksa*, were transiting the Kattegat off the Danish coast on 22 November 2004 when they collided. Both were following the recommended route 'T' on a south-easterly course. The weather was overcast and windy with rain, localised sleet and poor to moderate visibility. It was still dark. *Cepheus J* was proceeding at 16 knots, and *Ileksa* at 6.5 knots. VDR records from *Cepheus J* were recovered and analysed during the subsequent MAIB investigation.

On *Cepheus J*'s bridge, the chief officer had sent the lookout to clean the crew mess room, while he continued completing paperwork, standing at the chart table on the port side of the bridge. From there, he had an unrestricted view ahead, and close by, to his right, the displays of the ECDIS and radar were available. He did not see *Ileksa* until after the collision.

On board *Ileksa*, the third officer had just taken the watch and the master was also on the bridge, sending a daily report to the company. *Cepheus J* was noted astern by radar at between 3 and 3.5 miles and visually sighted astern at 1.5 miles. When the ships were 0.5 mile apart, *Ileksa* called *Cepheus J* by VHF radio to establish what her intentions were. When no reply was heard, and with the ships approximately 0.3 mile apart, *Ileksa* began to take evasive action, but with the wind ahead was not able to turn sufficiently to avoid collision.

At 0519 UTC, the two ships collided, with *Cepheus J*'s bow striking the stern of *Ileksa*. The impact caused severe damage to *Ileksa*'s stern and holed *Cepheus J* above the waterline. Both vessels were able to resume their voyages; there were no injuries and no pollution.

The accident was investigated by the UK's Marine Accident Investigation Branch and the Malta Maritime Authority, as a joint investigation under the International Maritime Organization's (IMO's) Code for the investigation of marine casualties and incidents.

Factors identified during the joint investigation were: the officer of the watch (OOV) on board *Cepheus J* being distracted by radio news and paperwork; the absence of a lookout on board *Cepheus J*; the poor use of Automatic Radar Plotting Aid (ARPA) and Automatic Information System (AIS); poor application of the COLREGS and both ships following the same ground track.

Subsequent to the accident, the operating company for *Cepheus J* has addressed the issues raised, and the Malta Maritime Authority has issued advice to the managers of *Ileska*.

Recommendations have been made to the International Chamber of Shipping to highlight the need for increased vigilance when transiting IMO recommended routes, and also to advise shipping companies on the value of VDR replay to confirm that company instructions are carried out at sea.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF *CEPHEUS J* AND *ILEKSA*, AND THE ACCIDENT

Vessel details		<i>Cepheus J</i>
Registered owner	:	Mare Schiffahrtsges.mbh&Co.KGms Cepheus J
Manager(s)	:	Jüngerhans Management Services GmbH &Co KG
Port of registry	:	London
Flag	:	UK
Type	:	Container ship
Built	:	Detlef Hegemann Rolandwerft, 2003
Classification society	:	Germanischer Lloyd
Construction	:	Steel – Ice class E3
Length overall	:	133.6m
Gross tonnage	:	6454
Engine power and/or type	:	MAK 8M43 7200kW
Service speed	:	16.5 knots
Other relevant info	:	Single screw, controllable pitch propeller, single rudder, bow thrusters
Persons on board	:	13
Injuries/fatalities	:	None
Damage	:	Hole in bow, above waterline. Plating dented along starboard shoulder

Vessel details***Ileksa***

Registered owner	:	Ileksa Shipping, Valletta, Malta
Manager(s)	:	INOK N V, Verbindgsdok, Oostkaai 5-7 Antwerp
Port of registry	:	Valletta
Flag	:	Malta
Type	:	General cargo
Built	:	Sudostroitelnyy Zavod, Novogorod 1996
Classification society	:	Russian Maritime Register of Shipping
Construction	:	Steel
Length overall	:	140m
Gross tonnage	:	4955
Engine power and/or type	:	2 x 8NVDS-48A-3U 2640 hp/ 1940kW
Speed - full load	:	10.5 knots
Other relevant info	:	Twin screw, single centreline rudder. Screw in nozzles
Persons on board	:	15
Injuries/fatalities	:	None
Damage	:	Extensive damage to the stern, including a split in the aft ballast tank, deck distortion, damage to the stern anchor and denting of port quarter plating.

Accident details

Time and date	:	0519 UTC on 22 November 2004
Location of incident	:	57°13.'85N 011°32.'65E 28 miles south-west of Gotenburg
Time of morning nautical twilight	:	0538 UTC
Time of sunrise	:	0711 UTC

1.2 BACKGROUND

Route T was established in 1971 by the Danish Government, with the objective of “ensuring the safety of navigation of large ships passing through Danish waters and also of reducing the risk of oil pollution resulting from the grounding and collision of tankers”. This is marked on the chart as a recommended route following the IMO guidelines (see Figure 3).

1.3 NARRATIVE

(Times given are UTC, since both ships kept different time zones)

lleksa sailed from Bilbao, Spain on 15 November 2004. She was in ballast and bound for Muuga in Estonia to load fertilizer. The passage across the Bay of Biscay and through the English Channel had been uneventful, with the three watchkeeping officers maintaining a 4 on / 8 off watch system. The weather had been rough crossing Biscay, and the ship’s service speed of 9.5 knots in ballast had rarely been achieved. As she was in ballast, the vessel tended to slam, especially in head seas, and this sometimes made it difficult for the crew to sleep. However, since entering the English Channel, the weather had been astern or on the beam, and sleeping had not been a problem.

At 0100 on 22 November, the chief officer took over the watch. The engines were set at full ahead, but the ship was only making 6 to 6.5 knots due to headwind and sea. The watch was again uneventful, with the ship entering the Kattegat and following route T. The passage plan had been drawn with the ship’s track 1 mile to the west of the Route T marked on the chart. The radar was in use and on the 6-mile range scale, with the display off-centred to the north-west, to maximise the view ahead. The view astern had been reduced to 3.5 miles, although the range scale was intermittently increased to monitor shipping at greater distances. The visibility had been reduced from time to time by passing showers of rain and sleet.

At about 0500, *Cepheus J* was detected astern by radar, and seen to be overtaking. Information concerning this ship was obtained from the AIS receiver, which was located on the port side of *lleksa*’s bridge.

Shortly before 0500 (0800 Moscow time), the third officer relieved the chief officer as OOW. The master was also on the bridge, preparing to send the daily telex to the company. As part of the handover, *Cepheus J* was pointed out astern as an overtaking vessel. The chief officer left the bridge. The ship was heading 165° and making good a track of 160° due to the weather conditions, at a speed of between 6 and 6.5 knots.

Cepheus J had left Rotterdam on 20 November, bound for St. Petersburg. The navigational watches were shared between the master, chief and second officers, following a 4 on / 8 off watch system. The cargo for this voyage included 61 refrigerated containers, fewer than the usual 70-100 carried on most voyages. The carriage of refrigerated containers required a record to be kept, at

6-hourly intervals, of the temperatures of each container. This was the chief officer's responsibility, and the maintenance of the records was time consuming. Collection of the temperature data from the refrigerated containers was obtained by the on-watch AB.

The chief officer took over the watch from the second officer at 0300. Visibility was poor, it was dark and an AB was on watch as lookout. The vessel was following Route T through the Kattegat at 16.5 knots, with the autopilot engaged. There were no defects with the bridge equipment, and the AIS information was displayed on the radar screen. Shortly after taking over the watch, the chief officer sent the AB to log the cargo temperatures, and then began working on paperwork. He was working at the chart table on the port side of the bridge.

At 0400, the AB went to carry out rounds of the ship. After returning to the bridge, he was sent away to sweep the stairs and tidy the crew mess room. The chief officer continued to work on his paperwork at the chart table and, occasionally, at a computer situated on the starboard side of the bridge.

Just after 0500, the chief officer tuned the radio on the bridge to a station transmitting the news in Russian. The major item of news concerned the political situation leading to the presidential elections in the Ukraine.

Cepheus J was maintaining between 15.5 and 16.5 knots on a heading of 160°. The vessel was following the planned track 1 mile to the west of the Route T displayed on the chart. A VHF radio was on and tuned to channel 16.

At 0516, the OOW of *Ileksa* called *Cepheus J* on VHF channel 16 to ask what her intentions were. At that stage, *Cepheus J* was approximately 0.7 mile astern of *Ileksa*, and just on her starboard quarter. There being no response from *Cepheus J* to this and subsequent VHF calls, *Ileksa*'s master had the helm put into manual and altered hard to starboard. Because of the effect of the wind and weather, *Ileksa* did not alter her heading over the next few minutes. *Ileksa*'s master then decided to remain on his course and take any impact at the stern, rather than continue to try to turn and possibly expose the structurally weaker ship's side to a collision. At 0519, the ships collided, with *Cepheus J*'s bow striking *Ileksa*'s stern. The bow of *Ileksa* was pushed round to starboard through between 60° and 90°, and then altered rapidly back to port as *Cepheus J* passed her port quarter.

Most of the crew on board *Ileksa* were awake when the vessels collided, as it was 8.20am ship's time. After the collision, the general alarm was not sounded, and the internal communications were used to ensure that the vessel was checked for damage. The general alarm was not sounded on *Cepheus J* either, and because a different time zone was kept on board, most of the crew were still in bed. An AB, who acted as carpenter, was telephoned and instructed to check the forward spaces. Once he had reported back to the bridge that there was a hole in the bow, the rest of the crew were awoken by the AB on watch and set to work checking for water ingress and for other damage.

The manoeuvring data for *Ileksa* showed that, in calm conditions at full speed of 10.5 knots, with the vessel in ballast, a turn through 90° would have taken 1 minute and 5 seconds, with an advance of 2.0 cables and a transfer of 0.9 cables (**see Annex 1**). As can be seen from the diagram, for the first 1.3 cables of the advance, there is no transfer. The alteration of course to this point would be about 45°, taking about 40 seconds.

1.10 DISTRACTIONS

The chief officer on board *Cepheus J* was required to compile a record of the temperatures maintained in the refrigerated cargo containers. This required that the temperature be recorded every 6 hours. The method adopted was for the temperature to be obtained by the AB on watch and then transferred to a fair copy by the chief officer. There were 61 refrigerated containers on board *Cepheus J* at the time of the collision, which was fewer than usual, however, the chief officer still spent a considerable amount of time each day completing the fair copy temperature log. It was this task that the chief officer was engaged in at the time of the collision.

As a Ukrainian national, the chief officer was naturally interested in the political situation of his country in the run up to the presidential elections. This is why he chose to listen to the radio during the time he spent on watch.

1.11 THE INTERNATIONAL REGULATIONS FOR PREVENTING COLLISIONS AT SEA

An extract of the pertinent regulations which were a factor in this accident, are shown at **Annex 2**.

1.12 LOOKOUT

The regulations concerning lookout are given in the International Regulations for Preventing Collisions at Sea (COLREGS) Rule 5 (see Section 1.10), and in the Code for Standards of Training and Certification of Watchkeepers 1995 (STCW95). The UK MCA gives further advice in the form of Merchant Shipping Notices (MGN 137 M+F, MGN 202 M+F).

STCW 95 Section A VIII/2 part 3.1 describes the principles to be observed in keeping a navigational watch.

Section 12 states the responsibility of the OOW for safe navigation of the vessel, and compliance with the COLREGS. Section 13 reinforces the requirements of Rule 5 of the COLREGS, and further explains that the lookout shall serve the purpose of:

1. *Maintaining a continuous state of vigilance by sight and hearing as well as by all available means, with regard to any significant change in the operating environment;*

2. *Fully appraising the situation and the risk of collision, stranding and other dangers to navigation; and*
3. *Detecting ships or aircraft in distress, shipwrecked persons, wrecks, debris and other hazards to safe navigation.*

Section 14 requires that:

The look-out must be able to give full attention to the keeping of a proper lookout and no other duties shall be undertaken or assigned which could interfere with that task.

STCW95 also states that the management company of a ship also has a responsibility for ensuring that the obligations given in the Code are given 'full and complete effect'. It also reinforces the requirement for lookout given in Rule 5 of the COLREGS.

The International Safety Management (ISM) Code requires that a company's safety management system should ensure compliance with mandatory rules and regulations. This is achieved by, including in the safety management system, instructions and procedures to ensure the safe operation of ships and protection of the environment in compliance with relevant international and flag state legislation.

1.13 VHF RADIO

The VDR recording on *Cepheus J* included three audio channels. Channels one and two were general recordings from microphones positioned to port and starboard on the bridge. The third channel recorded the VHF channel in use on the ship. The VHF recorded was the set on the starboard side of the main console.

Both ships were fitted with AIS, so the names and call signs of both vessels were available to each other before the collision.

The recording of VHF transmissions obtained from the VDR shows that *Ileksa* did try to alert *Cepheus J*. The recorded transmissions before the collision were:

"What are you doing? Alter your course"

followed by:

"Red to Green, Red to green"

These transmissions were made in English, and further transmissions between the two vessels, after the collision, were in Russian, using ships' names during the call.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 FATIGUE

Fatigue was not an issue in this accident, with the OOWs on both ships recording sufficient hours of rest.

2.3 LOOKOUT

2.3.1 International Regulations for Preventing Collisions at Sea

Rule 5 of the above rules states:

“Every vessel shall at all times maintain a proper look-out by sight and hearing as well as by all available means appropriate to the prevailing circumstances and conditions so as to make a full appraisal of the situation and of the risk of collision.”

Cepheus J had a number of “available means appropriate”. They included: radar, AIS, VHF and a visual lookout. Employing any one of these would have alerted the OOW to the presence of *Ileksa*, and thereby enabled action to be taken to avoid collision.

For a period of at least 1 hour before the collision, the target of *Ileksa* was on the radar of *Cepheus J*, though appearing under the heading marker of the display (see **Figure 9**). This did not entirely hide the target, but did make it difficult to see. However, the AIS information was also being displayed on the radar, with the symbol for a ship underway appearing close to the west of the radar target, and clear of the heading line. This should have alerted the OOW to the presence of *Ileksa* ahead of him.

The VHF radio on *Cepheus J* was switched on and set to Channel 16. When *Ileksa* called to ask the intentions of *Cepheus J*, the call did not include the ship’s name, despite this being available to the officers on board *Ileksa*, from the AIS. Probably because of this, the call did not attract the attention of the OOW on *Cepheus J*, who, in any case, was listening to the news on another radio and carrying out paperwork.

With the visibility restricted to 1.5 miles, *Ileksa* would have been visible from *Cepheus J* for 9 minutes before the collision. *Ileksa* was displaying the lights required for a vessel of her size, which would have meant that *Cepheus J* would have seen at least a single white sternlight. However, additional lighting was illuminated around the accommodation, which would have enhanced the visibility of the vessel from astern.

An efficient lookout was maintained on *Ileksa*. *Cepheus J* was first noted astern by radar at 3.5 miles, visually sighted at 1.5 miles, and the ship's name and other details noted from the AIS.

2.3.2 STCW 95

The principles to be observed in keeping a navigational watch, as contained in STCW 95 Section A VIII/2 part 3.1, describes, in Section 12, the responsibility of the OOW for safe navigation of the vessel, and compliance with the COLREGS. Section 13 reinforces the requirements of Rule 5 of the COLREGS, and Section 14 emphasises that the sole duty of the lookout is the maintenance of a proper lookout.

The OOW had sent the dedicated lookout away from the bridge, to carry out cleaning tasks elsewhere in the ship. Therefore, he was acting as the sole lookout. Since it was still dark, this was in contravention of STCW 95, which states, in Section A VIII part 15, that *the officer in charge of the navigational watch may be the sole lookout in daylight*.

Cepheus J's OOW was involved in tasks which distracted him from his primary duty of lookout. This meant that he was not paying attention to the radar or to keeping a visual lookout. Because he was listening to the news on the radio, he was unable to monitor the VHF radio effectively so he missed yet another indication of the presence of *Ileksa*. Had the requirements of the STCW 95 Code been followed, an efficient lookout would have been maintained on *Cepheus J*, and it is unlikely that the collision would have occurred.

2.3.3 M Notice advice

M notices are issued in the UK by the MCA. They are designed to offer advice to both the merchant and fishing fleets, on all matters associated with the operation of ships at sea and in port. There are three categories of notice: Marine Guidance Note (MGN), Marine Information Note (MIN) and Merchant Shipping Notice (MSN).

M-notices in force at the time of the collision included:

MGN 137 (M+F) Lookout During Periods of Darkness and Restricted Visibility

MGN 202 (M+F) Navigation in Fog

MSN 1767(M) Hours of work, Safe Manning and Watchkeeping Provisions.

MGN 137 is a reminder to all UK ships, wherever they may be, and to other ships operating in UK territorial waters, of the legal requirements for keeping a proper lookout, especially during the hours of darkness. It draws the attention to the requirements of the STCW Code Section A-VIII.

MGN 202 discusses the requirements of the COLREGS, with particular emphasis on good watchkeeping practices. Although this collision did not occur in fog, the visibility was reduced, and the advice applies.

MSN 1767 reminds masters, owners and operators that it is not safe for the officer of the navigational watch to act as sole lookout during periods of darkness or restricted visibility.

It is clear, from the evidence collected during the joint investigations, that the watchkeeping arrangements in place on *Cepheus J* at the time of the collision, took no account of the advice contained in any of these M-notice.

2.3.4 International Safety Management Code (ISM)

The ISM Code expressly provides that it is the duty of the managing company to take all reasonable steps to ensure that the ship is operated in a safe manner. The company must have established and implemented an effective safety management system which includes procedures to ensure the safe operation of ships, as well as reporting accidents and non-conformities.

The managing companies of both *Cepheus J* and *Ileksa* had implemented the ISM Code on board their respective vessels, and issued comprehensive instructions for the maintenance of a proper and effective lookout, which closely followed the requirements of the STCW 95 Code.

The results of internal and external audits will usually provide managers with an assessment of how well its Safety Management System is working on board its vessels. The presence of company or external audits on board a vessel will often ensure that ship's staff are careful to simply be seen to comply with laid-down procedures and working routines. Evaluation of VDR data taken from vessels following accidents, has provided the MAIB with invaluable evidence on how ships normally operate away from the scrutiny of company officials or external auditors. As a consequence, the MAIB believes the routine examination of VDR data would provide ships' managers with an incontrovertible assessment on the standards of watchkeeping displayed by ships' staff under 'normal' operating conditions.

2.3.5 Application of the regulations

The managing companies of both vessels had followed the requirements of the ISM Code, and had issued instructions to the ships' staff concerning lookout and navigational watchkeeping. Both sets of instructions re-stated the requirements for lookout, paraphrasing STCW 95 and/or the COLREGS.

The obligations for lookout rest with the individual OOW, overseen by the master of the ship. The routines should be established by the management company, and be to the satisfaction of the flag State.

2.4 APPLICATION OF THE COLREGS

2.4.1 Actions by *Cepheus J*

When the vessels were in radar range, but not in sight of each other, it would have been prudent for *Cepheus J* to have maintained a radar watch. VDR recorded the fact that the radar was operating, but that no targets had been acquired with the ARPA. It also showed that no other systematic observation of detected objects was being carried out. Had *Ileksa* been observed, *Cepheus J* could have altered to port or starboard to avoid collision under Rule 19. When the vessels were 1.5 miles apart, they could have seen each other visually, and the Section II rules would have applied. As the overtaking vessel, under Rule 13, *Cepheus J* was again obliged to give way, and could have altered in either direction to avoid collision. A 20° alteration of course on first sighting *Ileksa* would have allowed a 0.8 mile passing distance.

The lack of a visual lookout or a radar watch, meant that the presence of *Ileksa* was not noted on board *Cepheus J*, so no avoiding action was taken.

2.4.2 Actions by *Ileksa*

The approach of *Cepheus J* was closely monitored by *Ileksa*'s OOW, and risk of collision was determined. However, *Ileksa* was a slow ship, so it was not unusual for her to be overtaken. In this respect, the OOW was used to vessels taking late avoiding action. He was, therefore, not unduly concerned that *Cepheus J* was approaching and, apparently, taking no action. At a range of 2 miles, *Ileksa* could have altered course to avoid collision under Rule 19 of the COLREGS, since *Cepheus J* could not be seen. Having sighted the lights of *Cepheus J*, *Ileksa* was obliged to maintain her course and speed, as required by Rule 17 (a)(i) (**see Annex 2**). When it became obvious that *Cepheus J* was taking insufficient action, *Ileksa* began to take action as allowed under Rule 17(a)(ii). However, because the strong wind was right ahead, and the vessel altered only slowly out of the wind, the action had not started to take effect before the vessels collided. There had been an attempt to contact *Cepheus J* by VHF, but this did not use the ship's name or call sign (see Section 1.13). Other methods may be used to attract the attention of an approaching vessel within the COLREGS, but these were not used or attempted.

In conclusion, *Cepheus J* should have kept clear of *Ileksa*, but was not keeping a lookout, so was unaware of her presence. *Ileksa*, once it became clear that *Cepheus J* was taking no action, should have taken avoiding action earlier, given the prevailing weather conditions.

2.5 MANOEUVRABILITY

As the design and build of *Ileksa* was for inland waters operation, the propulsion and steering arrangements fitted produced limited manoeuvring characteristics in open sea operation. Her manoeuvring data showed that, in calm conditions, at full speed of 10.5 knots, in ballast, a turn through 90° would have taken 1 minute and 5 seconds, with an advance of 2.0 cables and a transfer of 0.9 cables (**see Annex 1**). As can be seen from the diagram, for the first 1.3 cables of the advance, there is no transfer. The alteration of course to this point would be about 45°, taking about 40 seconds.

Because the vessel's accommodation was at the after end, this would have had the effect of a sail, and would have tended to keep her head to wind. Turning the vessel off the wind would have been difficult, especially with the winds of near gale force from ahead being experienced at the time of the collision. This situation was worsened by her limited manoeuvring characteristics. With the speed through the water reduced by the headwind, the water flow over the rudder was reduced. Since the single rudder was positioned on the centreline, it was clear of the propeller wash; this, too, reduced its efficiency.

In conclusion, *Ileksa*'s inherent poor manoeuvring characteristics, especially in the prevailing weather conditions, should have encouraged earlier action, or a more concerted attempt to contact *Cepheus J* by VHF radio.

2.6 PASSAGE PLANNING

The principles to be observed in voyage planning are contained in IMO Resolution A893(21). This includes advice that routing measures should be adhered to. Route T had been established in 1971 (see Section 1.3). As the main route from the North Sea to the Baltic Sea, it would have been the obvious choice for either ship for its intended voyage. The median line with direction arrows was shown on the charts in use on both vessels. This indicated that the median line should have been left to port, by both north and south-bound vessels. By leaving a distance between the planned track and the median line, the two opposing streams of traffic are separated, reducing the number of "head-on" encounters.

Both ships had picked a track 1 mile to the west of the median line, which was a reasonable choice. However, knowing that this was the main route through the area, should have encouraged heightened vigilance due to the increased likelihood of encountering other traffic heading in the same direction.

IMO resolution A893(21) also states that account should be taken of "volume of traffic likely to be encountered" when appraising the passage plan. The fact that the chosen track followed an IMO recommended route, should have indicated to the navigator that there might have been other traffic using it, therefore the lookout should have been enhanced.

2.7 USE OF RADAR

2.7.1 Radar on *lleksa*

The radars were provided with an elementary plotting aid, which required the operator to mark the position of the target on the screen at regular intervals. The computer then performed the calculations necessary to provide the observer with the required information about the target. Although a useful tool, it required continuous monitoring to ensure that the target was marked sufficiently frequently to give reliable and up to date information.

These radars were also capable of displaying both true and relative trails. At the time of the collision, the radar was displaying relative trails, and the plotting facility was not being used. However, a paper plot was being maintained by the third officer, from ranges and bearings read off from the radar by the master.

Not all the facilities of the radar were being used, but *lleksa* was aware of the presence of *Cepheus J* and had assessed the risk of collision.

2.7.2 Radar on *Cepheus J*

Cepheus J was fitted with a modern NACOS Integrated Bridge System designed by STN Atlas (now SAM Electronics). The two radar screens were independent and inter-switchable, with the port screen mounted next to the electronic chart display. The radars were both fitted with ARPA facilities.

The ARPA facility allows a target to be tracked after manual acquisition by the operator. It also enables a guard zone to be established around the ship. This guard zone facilitates the automatic acquisition of any target appearing within the zone. Should a target appear in the zone, it is automatically acquired and an alarm will sound to alert the operator. Once the target is acquired, it will be tracked by the ARPA, and information concerning its track will be calculated. This includes its closest point of approach and the time at which this will occur. An alarm facility is available, which will warn the operator if the target will breach user defined limits for CPA and TCPA.

The VDR recording shows that no targets had been acquired and that no guard zone had been established. Had either manual or automatic acquisition of targets been used, the OOW would have been alerted to the possible collision with *lleksa*, and could have taken action to allow safe passing.

In addition to the ARPA facility, the radar displayed AIS information. This put a triangular symbol on the screen adjacent to the position of the target. Although this did not coincide exactly with the radar target, there could be no doubt that the two were the same vessel, since there were no other targets in the vicinity. As previously stated, the radar return of *lleksa* was positioned underneath the heading marker of *Cepheus J*'s radar, rendering it difficult to see. The AIS target positioned to one side of the heading marker, would have highlighted the presence of a target near that position.

Relative trails were selected for display at the time of the collision. Since the relative approach of *Ileksa* was down the heading marker, the relative trails would also have painted there, so would have been masked to the observer.

The OOW was not using the integrated bridge equipment, with which he was provided, to its best effect.

2.8 USE OF VHF RADIO AND OTHER SIGNALS TO ATTRACT ATTENTION

2.8.1 VHF radio

When *Ileksa* called *Cepheus J* before the collision, no ships' names or call signs were used during either of the two transmissions. The sounds recorded by the VDR, on *Cepheus J*'s bridge, show that these transmissions were clear. They would not, however, have been sufficiently different to other random transmissions picked up during the minutes before the collision, for the OOW's attention to have been distracted from the radio news (**see Figure 10**).

The MAIB believes that strict adherence to the ColRegs will always prevent collision, and the use of VHF as a means of negotiating avoiding action at sea can lead to confusion and should be avoided.

2.8.2 Other signals to attract attention

Apart from the use of VHF, no signals were attempted by *Ileksa* to attract the attention of *Cepheus J*. Additional signals are permitted under the COLREGS (Rule 36). It is likely that one of the most efficient in this case would have been to have pointed a searchlight in the direction of *Cepheus J*.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES

The following safety issues have been noted during the investigation. They are not listed in any order of priority.

1. The lookout on the bridge of *Cepheus J* had been sent to carry out cleaning duties elsewhere on the ship. [2.3.2]
2. An efficient lookout was maintained by *Ileksa*. [2.3.1]
3. *Cepheus J*'s OOW was involved in tasks that distracted him from his primary duty of lookout. [2.3.2]
4. The watchkeeping arrangement in place on *Cepheus J* at the time of the collision, took no account of the advice contained in M Notices. [2.3.3]
5. The management companies of both vessels had adopted the requirements of the ISM code, and issued instructions to the ship staff concerning lookout and navigational watchkeeping. [2.3.5]
6. *Cepheus J* was not keeping a lookout. This meant that the presence of *Ileksa* was not detected, so no avoiding action was taken. [2.4.1]
7. Once it became clear that *Cepheus J* was taking no action, *Ileksa* should have taken action earlier, given the prevailing weather conditions. [2.4.2]
8. The poor manoeuvring characteristics of *Ileksa*, in the prevailing weather conditions, should have encouraged earlier action. [2.5]
9. The chosen track followed an IMO recommended route. This should have indicated to the navigator that there might have been other traffic using it, and prompted a heightened lookout. [2.6]
10. Not all the facilities of the radar on *Ileksa* were being used, but the presence of *Cepheus J* was known and the risk of collision had been assessed. [2.7.1]
11. *Cepheus J*'s OOW was not using the integrated bridge equipment, with which he was provided, to its best effect. [2.7.2]
12. *Ileksa* could have used signals permitted in the COLREGS which might have attracted *Cepheus J*'s attention in sufficient time to have allowed action to be taken to avoid collision. [2.8]